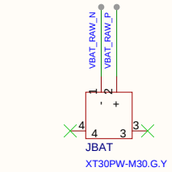
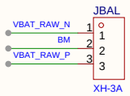
DesktopAssistant Integrated | Battery protection and balance / 电池保护与均衡

Revision 1.2 - 20260912 01:05

2S battery input / 原始电芯端

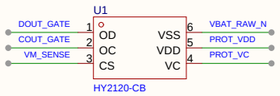




JBAL: 1=B , 2=BM, 3=B+; verify cable polarity.

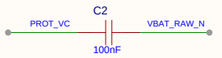
HY2120-CB protection / 过充过放与短路保护

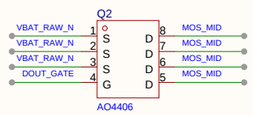


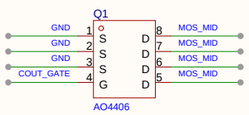


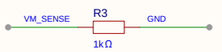






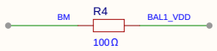




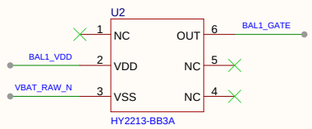


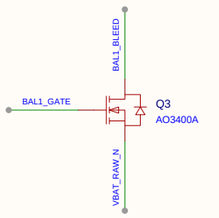
B → Q2(S-D) → MOS_MID → Q1(D-S) → protected GND

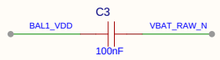
Lower cell balance / 下节电芯

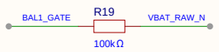




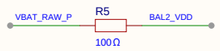




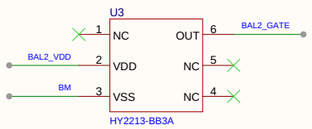


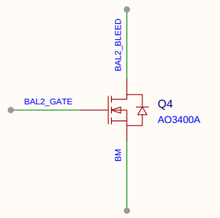


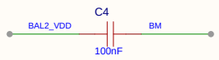
Upper cell balance / 上节电芯

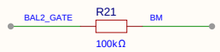












Balance current ≈ 19mA at 4.2V; each bleed resistor ≈ 0.08W.

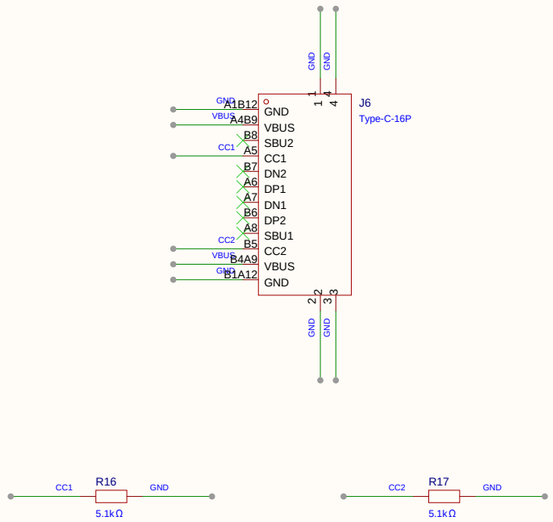
GND is protected pack negative. Do not short GND to VBAT_RAW_N.

Protection and balance thresholds require bench validation before battery use.

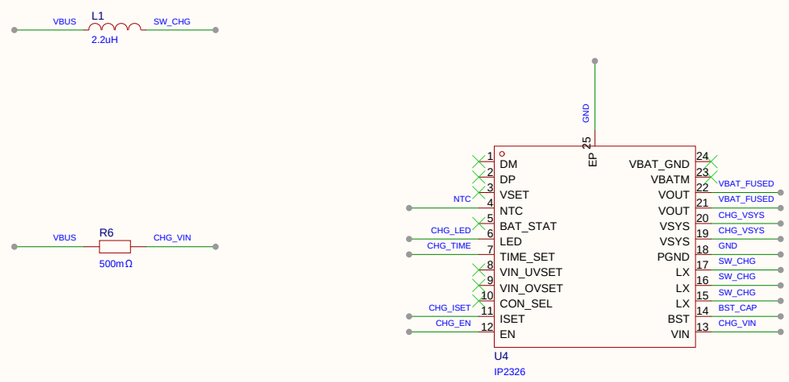
DesktopAssistant Integrated | USB-C charger / 2S 锂电充电

Revision 1.2 - 2026-09-22 2025-05

5V USB-C input / 专用 5V 适配器



IP2326 synchronous boost charger



Input / output decoupling



Charge enable and status / 默认禁止充电



Charge settings / 电流、超时和电池温度



2S: CON_SEL NC; 8.4V: VSET NC; VIN UV/OV: NC defaults.

External HY2213 balance: IP2326 pins 23/24 left NC.

Use regulated 5V ≥ 2A adapter. D+/D unused; no QC request.

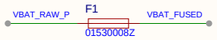
JP_CHG defaults OPEN. NTC must touch battery; charge with main switch OFF.

0.353A is calculated; validate charge current/termination before battery use.

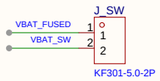
DesktopAssistant Integrated | Power and MCU / 配电与主控

Revision 1.2 - 2026-09-12 03:05 / 05

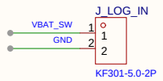
Protected pack → fuse → switch → regulators

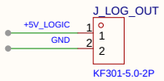


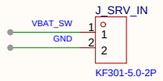
Fit 029707.5WXNV 7.5A fuse into F1.

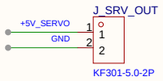


1=fused input; 2=switched output.



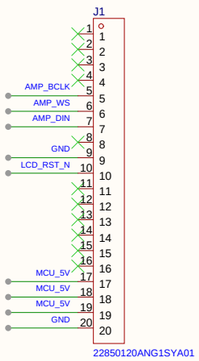


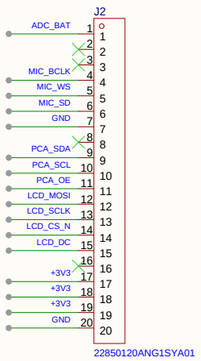




Two external regulator modules; outputs are separate. Set both to 5.0V before connection.

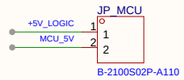
ESP32-S3 carrier sockets / 保留已核对的开发板针序





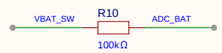
MCU 5V isolated by JP_MCU; OPEN during USB debugging.

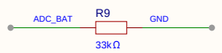
Logic supply / USB isolation

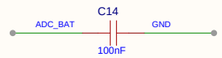




Battery ADC / switched measurement







ADC_BAT ≈ 2.08V at pack 8.4V.

Each module connector: pin 1 positive, pin 2 GND. Modules are external, not on-board footprints.

J_SW is a cable interface, not a diode. Use a DC-rated switch appropriate for servo startup current.

DesktopAssistant Integrated ||Display, audio and UI 显示与音频显示音频与交互

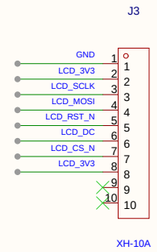
Revision 1.2 - 2026-04-05 / 2026-04-05

3.3V LCD supply / 独立隔离的屏幕支路



Select ONE LCD source. Keep JP_LCD open when J10 is used.

SPI display harness / 屏幕线束接口



J3: cable to external HS24S010B pins 1-10; pins 9/10 NC.

I2S module harnesses / 音频线束接口



J4: 1=BCLK, 2=WS, 3=SD, 4=3V3, 5=GND, 6=L/R GND.

J5: 1=BCLK, 2=WS, 3=DIN, 4=5V, 5=GND, 6=GND aux.

These are wire-harness pinouts; follow signal names on the modules.

Speaker connects to amplifier OUT+/OUT , not motherboard GND.

HS24S010B interface notes / 屏幕接口说明

ST7789V2 - 240 x 320 - 4-wire SPI - 3.3V only

CN1: 1 GND | 2 VCC | 3 SCL | 4 SDA | 5 RES

CN1: 6 DC | 7 CS | 8 BLK | 9 FSO | 10 FCS

BLK is a control input: held HIGH for backlight ON.

FSO/FCS: optional font/flash; pins 9/10 are NC here.

Pin table and mechanical CN1 agree; inset schematic differs.

Check actual screen silk before wiring the XH-to-2.54mm cable.

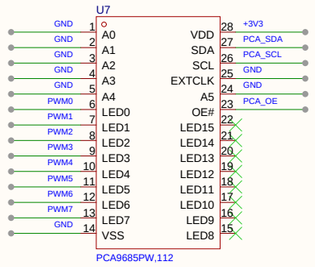
All ESP32 signals are 3.3V. Do not connect 5V logic directly to inputs.

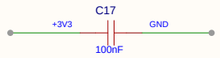
LCD source follows RevB isolation strategy. This page does not connect the two regulator outputs.

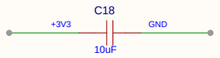
DesktopAssistant Integrated | PWM and servos / 舵机控制与配电

Revision 1.2 - 2026-09-12 05:05 / 05

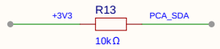
PCA9685 " I2C 0x40 " OE high = outputs disabled

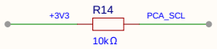


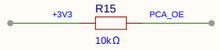




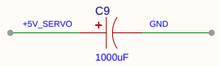
I2C pull-ups and safe startup

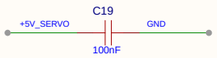




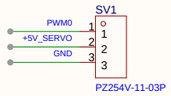


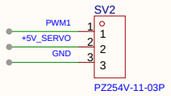
OE defaults HIGH. Firmware must initialize PWM before enabling.

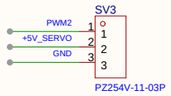


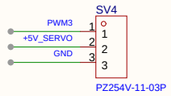


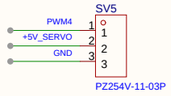
Eight 3-pin servo headers / 1=PWM, 2=+5V_SERVO, 3=GND

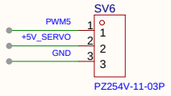


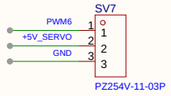


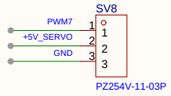












Servo power has its own 5V converter; route high-current return directly to protected power ground.

LED8-LED15 unused. EXTCLK tied to GND for internal oscillator. All address pins are GND.

No servo motion until supply, polarity, current limit and firmware fail-safe behavior have been tested.